

Initial Data Assessment: Distributions of Quarterly Changes by Bank Size, 1992–2026

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Abstract

This document assesses how the cross-sectional distributions of four bank outcomes—total assets, the equity-to-assets ratio, return on assets (ROA), and net interest margin (NIM)—change over one-, two-, and three-quarter lags, separately for the largest 1% of FDIC-insured institutions and for all other banks. The main finding is that different variables, time-periods, and sizes of banks exhibit different distributional features. The next pressing question is how to appropriately model these distributions in a way that is consistent with the data and useful for macro-financial analysis. The later section of this document will propose an MLE approach.

1 Data and Construction

The sample is the FDIC Statistics on Depository Institutions bank-quarter panel from 1992Q1 to 2026Q2, with 1,127,856 bank-quarters and 17,121 institutions identified by FDIC certificate number. Insured U.S. branches of foreign banks (FDIC charter class OI; 40 institutions, 2,069 bank-quarters) are excluded: they report assets but no equity or income, and the FDIC records their ROA and NIM as zero, which would otherwise enter the distributions as spurious zero changes. In each quarter t , banks are ranked by total assets. The top 1% form the *large-bank* group (between 42 and 143 institutions, depending on the year) and the remaining 99% form the *other-bank* group. Important note, for the actual analysis portion, the 1% value will likely have to be extended to 5% or 10% to satisfy the data needs of both bootstrap resampling or maximum likelihood estimation. Group membership is re-assigned every quarter.

For bank i and horizon $k \in \{1, 2, 3\}$, the outcome is the change from quarter $t - k$ to quarter t , matched on the same institution:

$$\begin{aligned}\Delta_k A_{it} &= 100 \times (A_{it}/A_{i,t-k} - 1) && \text{(total assets, percent)} \\ \Delta_k x_{it} &= x_{it} - x_{i,t-k} && \text{(equity/assets, ROA, NIM; percentage points)}\end{aligned}$$

Asset changes are expressed in percent because dollar changes are not comparable across institutions whose balance sheets differ by several orders of magnitude.

The FDIC reports ROA and NIM year-to-date and annualized, so differencing them directly would compare a single quarter with a full year every first quarter. Each bank's single-quarter ratio is therefore recovered first:

$$r_Q^q = Q \cdot r_Q^{\text{ytd}} - (Q - 1) \cdot r_{Q-1}^{\text{ytd}}, \quad r_1^q = r_1^{\text{ytd}},$$

within each calendar year, where Q is the quarter of the year. The formula is exact when average assets are stable within the year and a close approximation otherwise.

Figures 1–4 are fan charts. In each quarter, the line marks the median change across banks in the group, the dark band the 25th–75th percentiles, and the light band the 10th–90th percentiles. Rows correspond to the horizon k , and all six panels in a figure share one vertical axis. Gray shading marks NBER recessions.

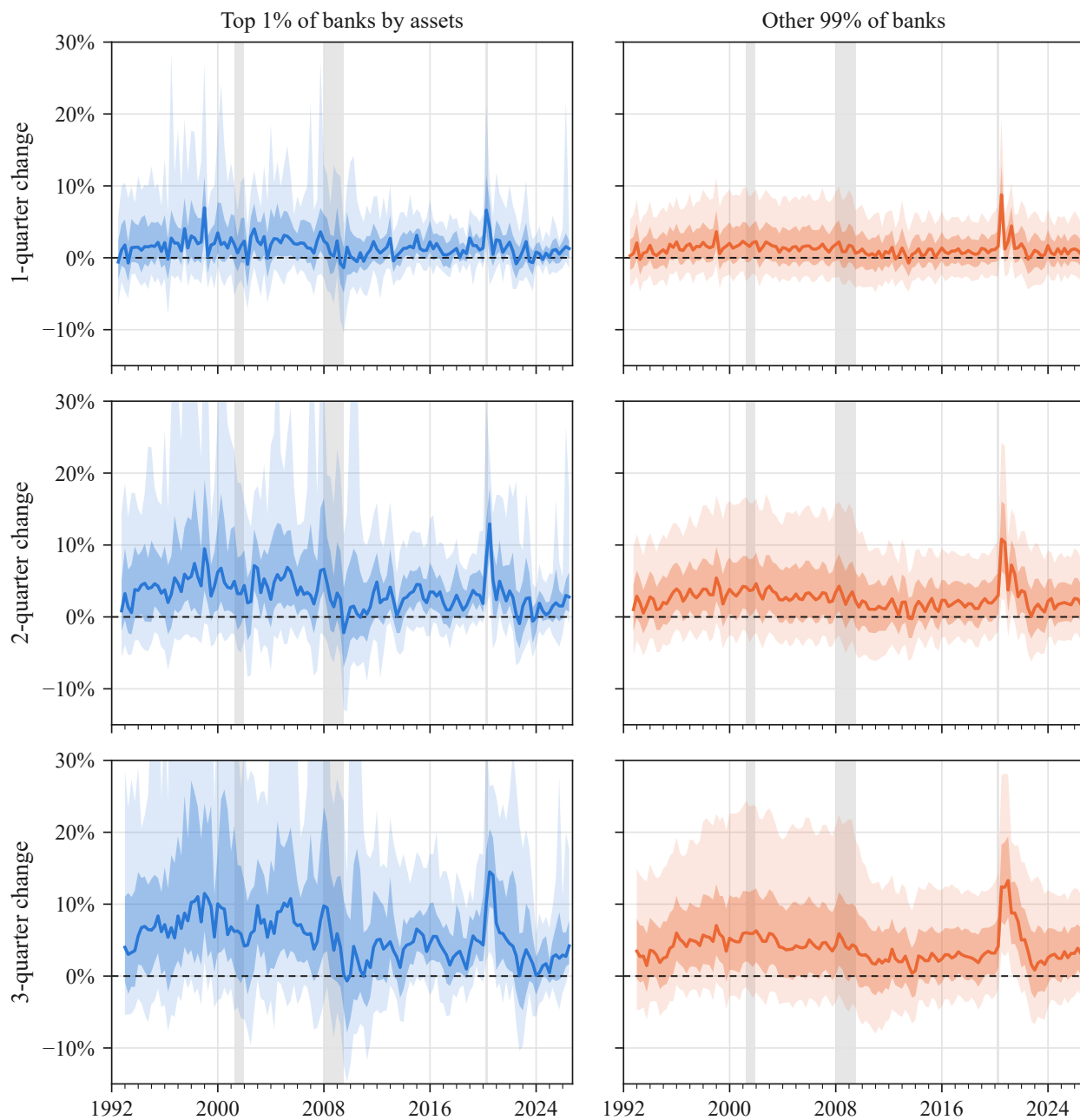


Figure 1. Change in total assets (percent).

Notes: Percent change in each bank's total assets from k quarters earlier. Changes include mergers and acquisitions. The vertical axis is capped at -15% and 30% ; the large-bank upper tail exceeds 100% in several late-1990s quarters. Line: median bank; dark band: 25th–75th percentiles; light band: 10th–90th percentiles. Shaded areas indicate NBER recessions. *Source:* FDIC Statistics on Depository Institutions; author's calculations.

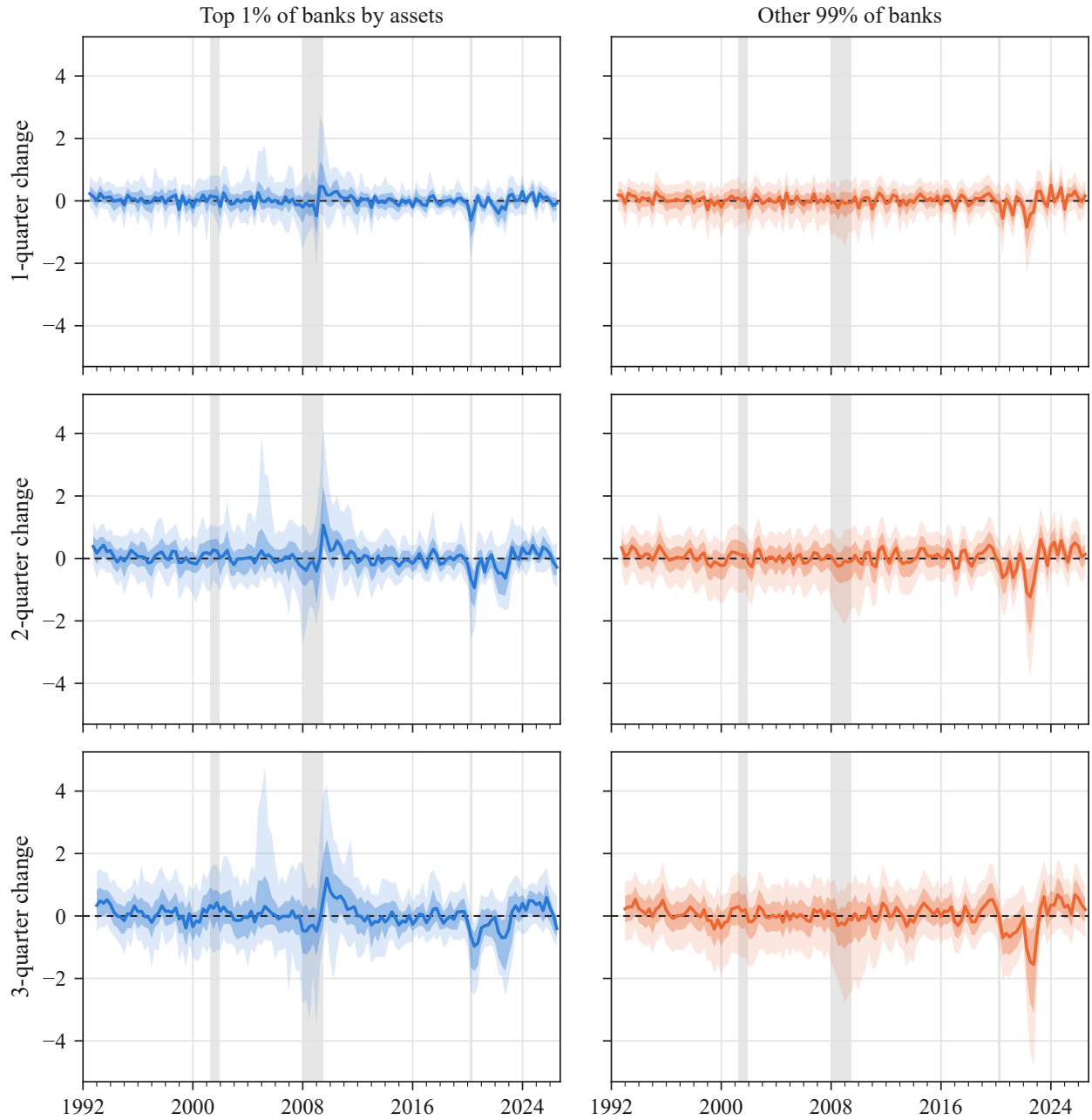


Figure 2. Change in equity capital to assets (percentage points).

Notes: Percentage-point change in each bank's ratio of total equity capital to total assets from k quarters earlier. Bands as in Figure 1. Shaded areas indicate NBER recessions. *Source:* FDIC Statistics on Depository Institutions; author's calculations.

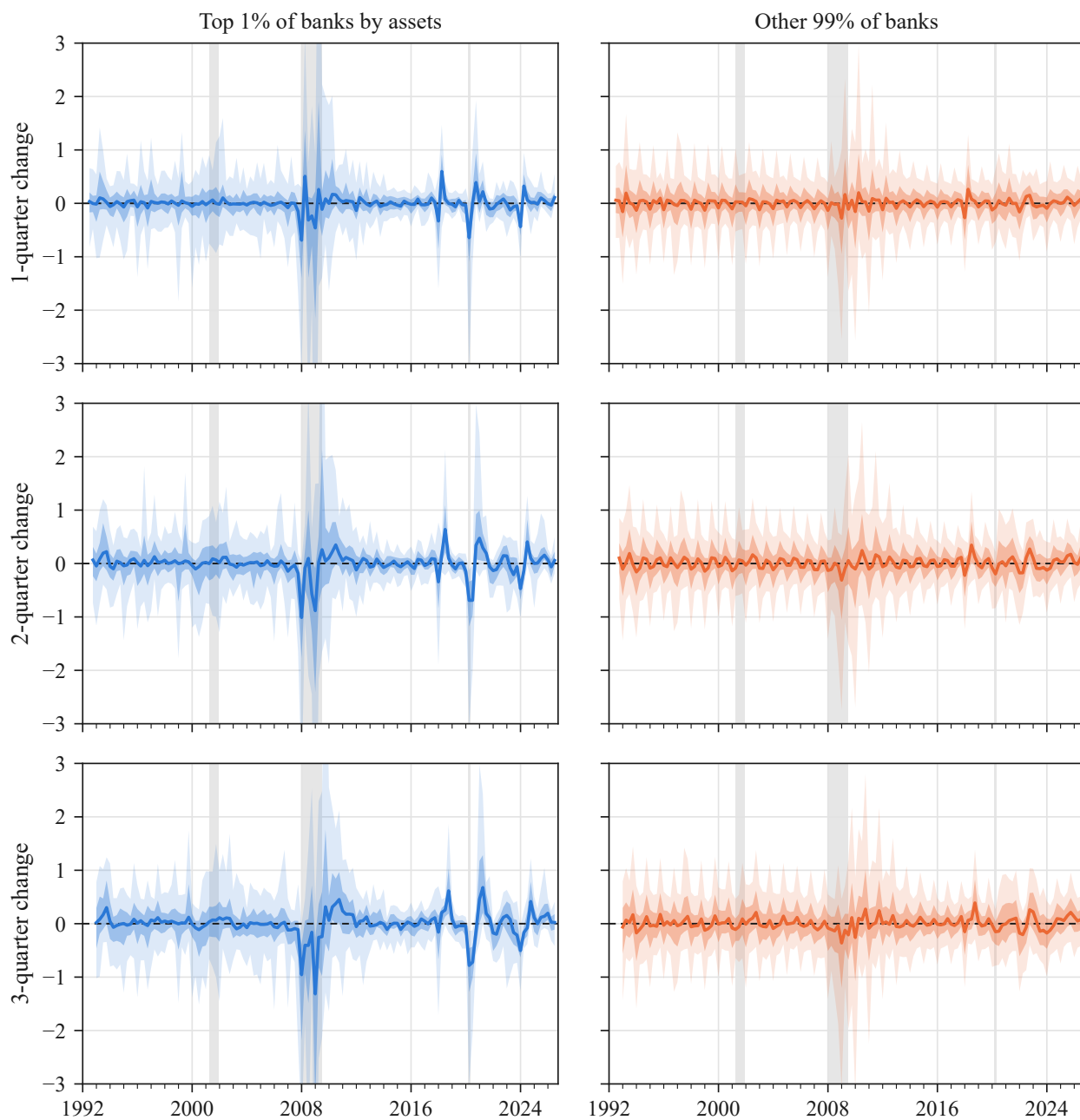


Figure 3. Change in return on assets (percentage points).

Notes: Percentage-point change in each bank's single-quarter, annualized ROA from k quarters earlier (see Section 1). The vertical axis is capped at ± 3 , which clips the large-bank tails in 2008–09. Bands as in Figure 1. Shaded areas indicate NBER recessions.
Source: FDIC Statistics on Depository Institutions; author's calculations.

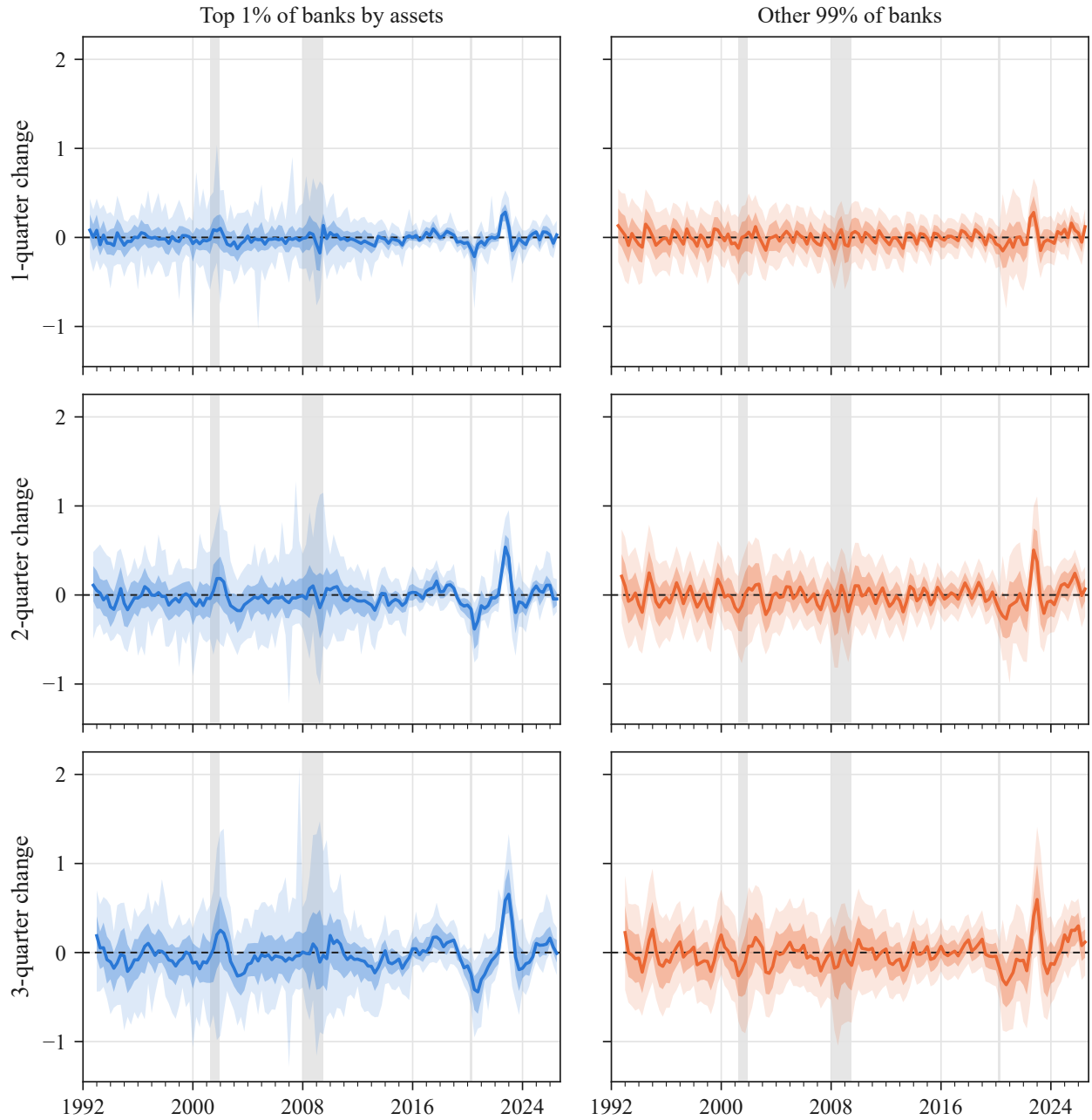


Figure 4. Change in net interest margin (percentage points).

Notes: Percentage-point change in each bank's single-quarter, annualized net interest margin from k quarters earlier. Bands as in Figure 1. Shaded areas indicate NBER recessions. *Source:* FDIC Statistics on Depository Institutions; author's calculations.

2 Visual Findings

Assets. Upon initial assessment, the different lags have similar distributions, just stretched out over the their tails. Additionally, the size of the upside tail for the top 1% of banks is much larger than the other banks, though the median seems somewhat stable. Also, the smaller asset-sized group appears to have a more symmetric distribution over the time period. Looking particularly during recessions, the downside tail of the large banks is much more pronounced than the other banks, which is consistent with the idea that larger banks are more exposed to macroeconomic shocks. This also mirrors trends in household income distributions.

Equity ratio. As opposed to asset distributions, equity ratio distributions are more symmetric and stable over time, with the exception of the 2008–09 financial crisis and COVID-19 recession. Both have overwhelmingly downside tails during recessionary periods, and appear to be left-skewed even in expansionary periods. The actual median does appear to be same across groups and moves similarly.

ROA. This measure appears to be the most volatile of the four, with the largest upside and downside tails. The large-bank group has a much more dense upside tail, especially during economic rebound periods. The furthest outliers of the majority of banks appears much more sparse. The sawtooth pattern of the right-hand group is likely due to the fact that ROA is annualized, and the first quarter of each year is compared to the fourth quarter of the previous year.

NIM. The net interest margin distributions appear the most stable of the four. The sawtooth pattern is again present due to the annualization. Distributions tend to be very tight to the median outside of recessionary periods. Overall, the effects seem to symmetric.

3 Potential Concerns

Grouping on current size selects on growth. Because group membership is determined by assets at t , a bank that crossed into the top 1% through rapid growth or an acquisition is, mechanically, a large bank with a large asset change. This inflates the upper tail and the right skew of large-bank asset growth, and to a lesser extent anything correlated with growth. A robustness check is to assign groups using assets at $t - k$, or to fix membership at the start of each episode.

Survivorship and entry. A change is defined only if the bank reports in both $t - k$ and t . Banks that fail or are absorbed drop out after their last filing, so the very worst outcomes, such as a bank’s final quarter before closure, can be missing from the lower tail. New charters enter only after k quarters. Both effects matter most in 2008–10, when a large number of institutions failed.

Small groups make tails noisy. The large-bank group contains 42–143 banks, so its 10th and 90th percentiles are set by roughly the 4th to 14th most extreme institution. This creates larger skewness artificially just by the small sample size, and it makes the tails of the large-bank distributions more volatile than those of the other-bank group. For the actual analysis, this will likely need to be extended significantly.

Few recession quarters, one dominant episode. Only 11 of 138 quarters overlap a recession, and 6 of them fall in 2008–09. The recession columns of the tables are therefore mostly a description of the financial crisis. The 2001 recession barely registers in the bank data, and the 2020 recession is only two quarters long and is confounded by pandemic-era deposit inflows and loan-loss provisioning. For this reason, finding some way to correlate the distributions with a macroeconomic indicator such as the unemployment rate or the federal funds rate may be more informative than a binary recession indicator. However, there is heavy endogeneity in the relationship between such outcomes and our data, so finding an exogenous measure or creating some limits on the predictive power of the macroeconomic indicator may be necessary.

Overlapping horizons and seasonality. It appears that distributions generally move in the same way over the span of a year, so $t - 1$, $t - 2$, and $t - 3$ are not independent. Particularly, it appears that abnormal changes in across a one quarter lag correlate to an even larger, same-signed change in the three quarter lag.

Axis caps. The vertical axes of Figures 1 and 3 are truncated to keep the medians and inner bands legible.

4 MLE Proposal

MLE Proposal. To study both how banks respond to macroeconomic shocks and how the cross-sectional distribution of those responses changes, I propose a heteroskedastic model solved by maximum likelihood. Unlike a standard linear regression, which recovers only the conditional mean response assuming bivariate normality, this approach jointly models the mean and the dispersion of bank outcomes, allowing both to depend on the macroeconomic shock and on bank characteristics. Let y_{it} denote some variable of interest for bank i in quarter t , s_t an identified macroeconomic shock, and w_{it} a vector of bank characteristics such as capitalization, liquidity, and size. The model is

$$y_{it} = \alpha + \beta s_t + w'_{it}\gamma + (s_t \cdot w_{it})'\theta + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, \sigma_{it}^2), \quad (1)$$

$$\ln \sigma_{it} = \delta_0 + \lambda s_t + w'_{it}\delta, \quad (2)$$

and the parameters are estimated by maximizing the log-likelihood

$$\ell(\alpha, \beta, \gamma, \theta, \delta_0, \lambda, \delta) = \sum_{i,t} \left[-\ln \sigma_{it} - \frac{1}{2} \ln(2\pi) - \frac{(y_{it} - \mu_{it})^2}{2\sigma_{it}^2} \right], \quad (3)$$

where μ_{it} is the conditional mean in equation (1). The coefficient β captures the average response to the shock, θ captures how that response varies with bank characteristics, and λ measures whether the shock widens ($\lambda > 0$) or compresses ($\lambda < 0$) the dispersion of bank responses. A likelihood ratio test of $\lambda = 0$ therefore provides a direct test of whether macroeconomic shocks alter the cross-sectional distribution of bank behavior.

Glaring Issues. First s_t needs to be exogenous to the bank outcomes, which is a strong assumption. An appropriate measure must be selected or some exogenous variable created via IV, for instance. Second, the model assumes that the conditional distribution of y_{it} is normal, and as normal distributions are symmetric, this still does not quite answer the question at hand, but provides a start that is relatively easy to implement.